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PHYSIOLOGY OF GIT ABSORPTION AND SELECTIVE MALABSORPTION

Basics of Absorption 00:01:30

STAGES OF ABSORPTION

1. Luminal : Enzymes mediated.
2. Small intestinal mucosal : Enterocyte.
   Villi → Brush borders contains enzymes.
3. Post mucosal : Mucosal epithelial cell → Into blood.

ANATOMY

Small intestine :

Proximal (2/5) : Maximum absorption (Jejunum > Duodenum).

Distal (3/5) : Absorption of Bile acid, Vit B_{12}, Mg^{2+} (Ileum).

Histology :

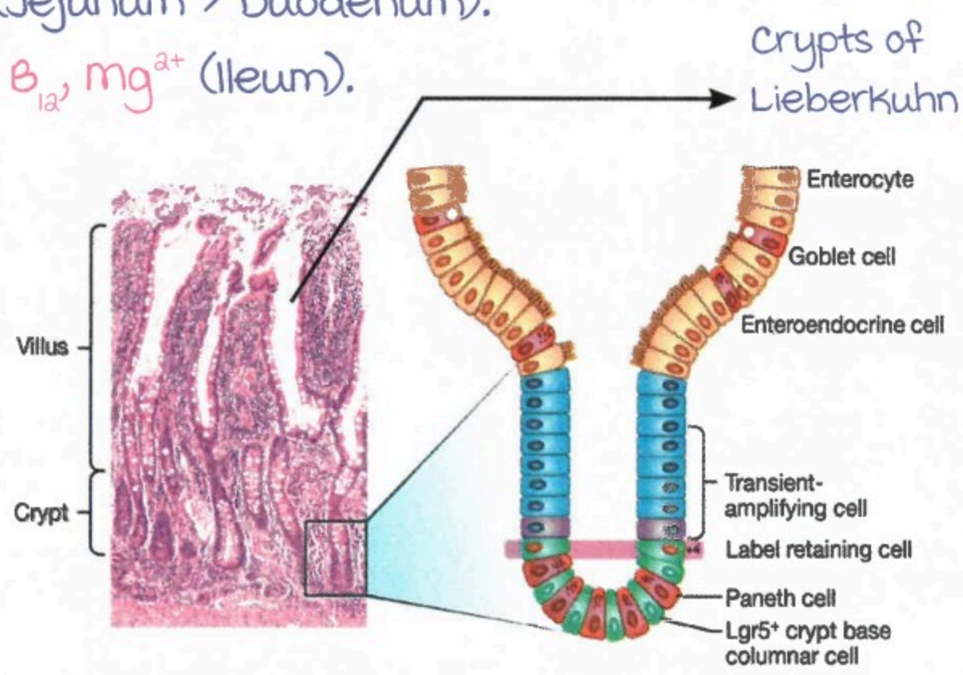
1. Mucosa :

a. Enterocyte :

- Villi (specific feature) :
  - Surrounded by crypts of Lieberkuhn.
  - Contains brush-border.
- Crypts :
  - Paneth cells (at base) : Produce immunological substances (alpha defensins, lysozymes, phospholipase A2).
  - Enteroendocrine cells.
  - Goblet cells.
- Continuous renewal : every 48-72hrs.

b. Lamina propria : Glands, blood vessels.

c. Muscularis mucosa : Tight attachment.



Small intestine mucosa with villi and crypts

Note :

Plica circularis : Visible mucosal folds on luminal surface of small intestine.

Interstitial cells of Cajal : Pacemaker cells of small intestine → Basal Electrical Rhythm.

2. Submucosa :
   - Brunner's glands (In duodenum) → Secretes bicarbonate (Alkaline).
   - Peyer's patches (In ileum).
     - Macroscopic lymphoid aggregates.
     - In typhoid : Hypertrophy.
3. Muscularis Propria.
4. Serosa.

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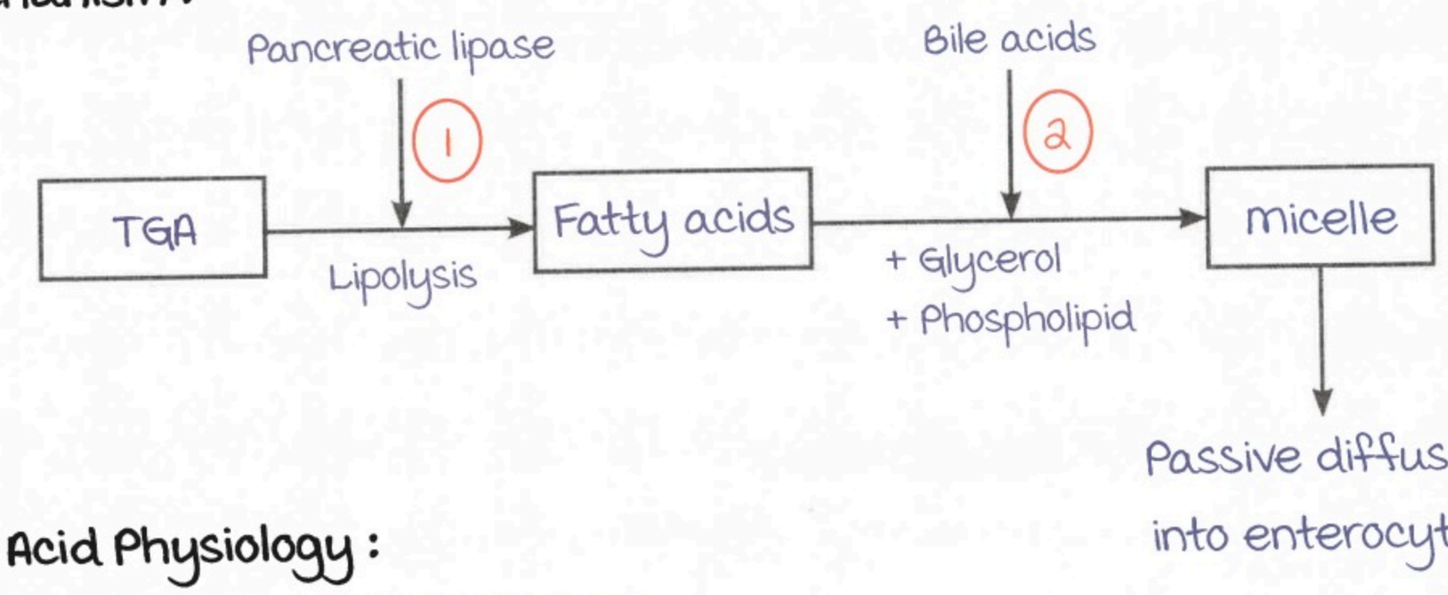
Peyer's patches on ileocolonoscopy

Fat Absorption

00:09:45

LUMINAL PHASE OF ABSORPTION

Mechanism :

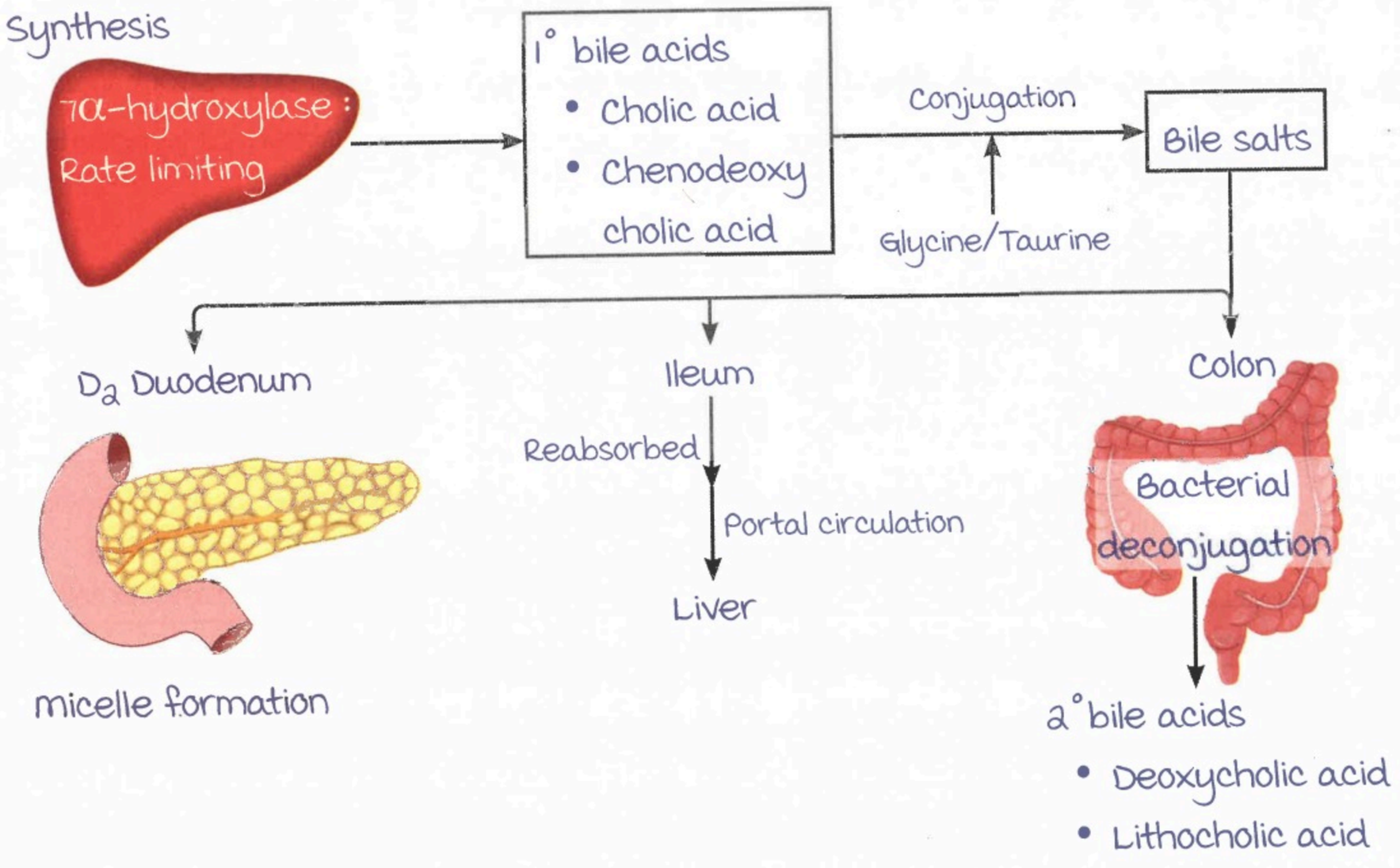


1. Chronic pancreatitis
2. Bile acid defect

Bile Acid Physiology :

- Formation & excretion : 500 mg.
- Body pool : 4g (maintained by enterohepatic circulation).

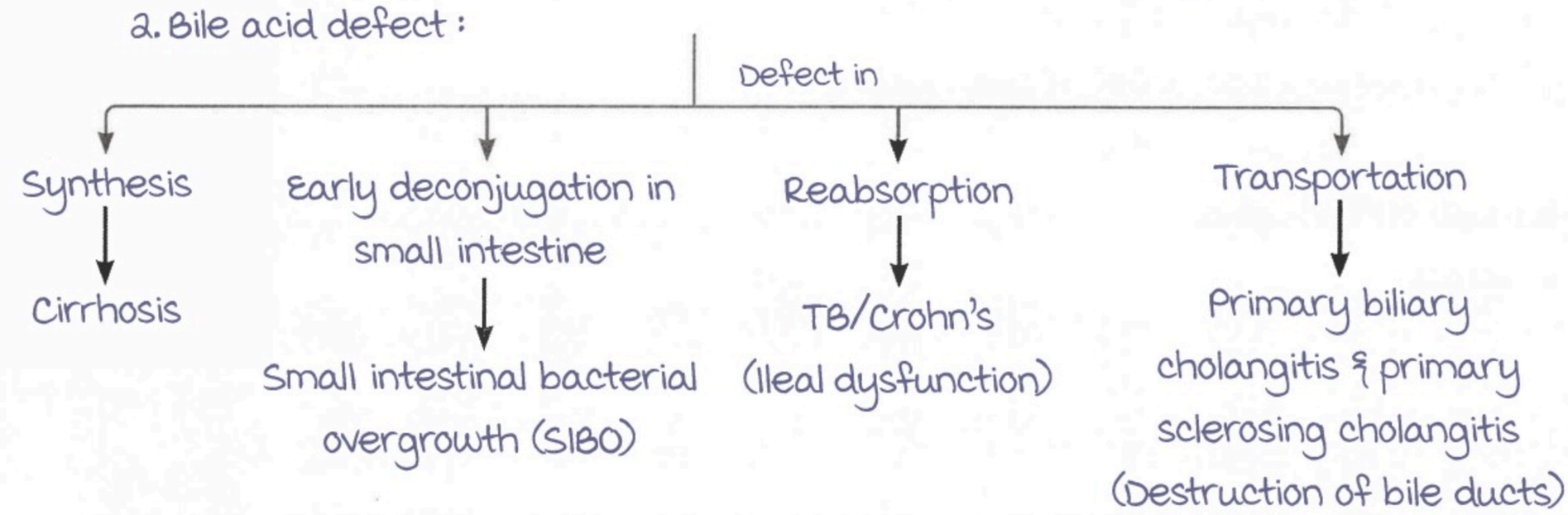
Enterohepatic circulation :



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Defects in luminal absorption :

1. Chronic pancreatitis → Affects pancreatic lipase → Lipolysis affected.
2. Bile acid defect :



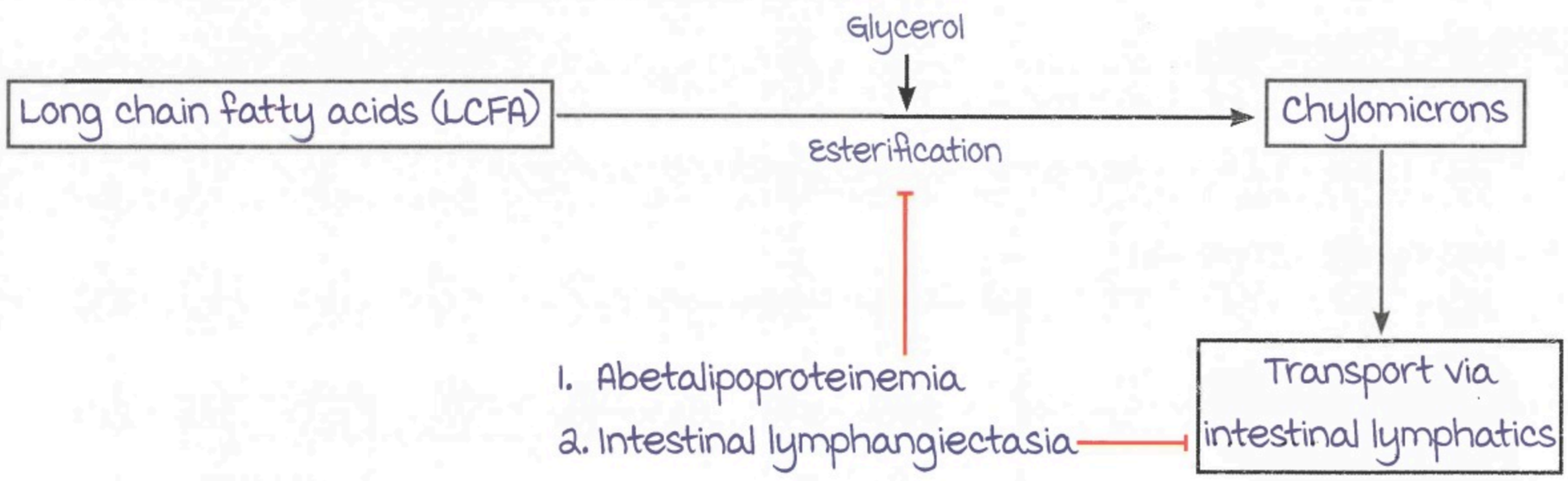
MUCOSAL PHASE OF ABSORPTION

Pathology : Defect in small intestinal mucosal integrity.

Etiology :

- Celiac disease.
- Whipple disease.
- Tropical sprue.

POST-MUCOSAL PHASE OF ABSORPTION



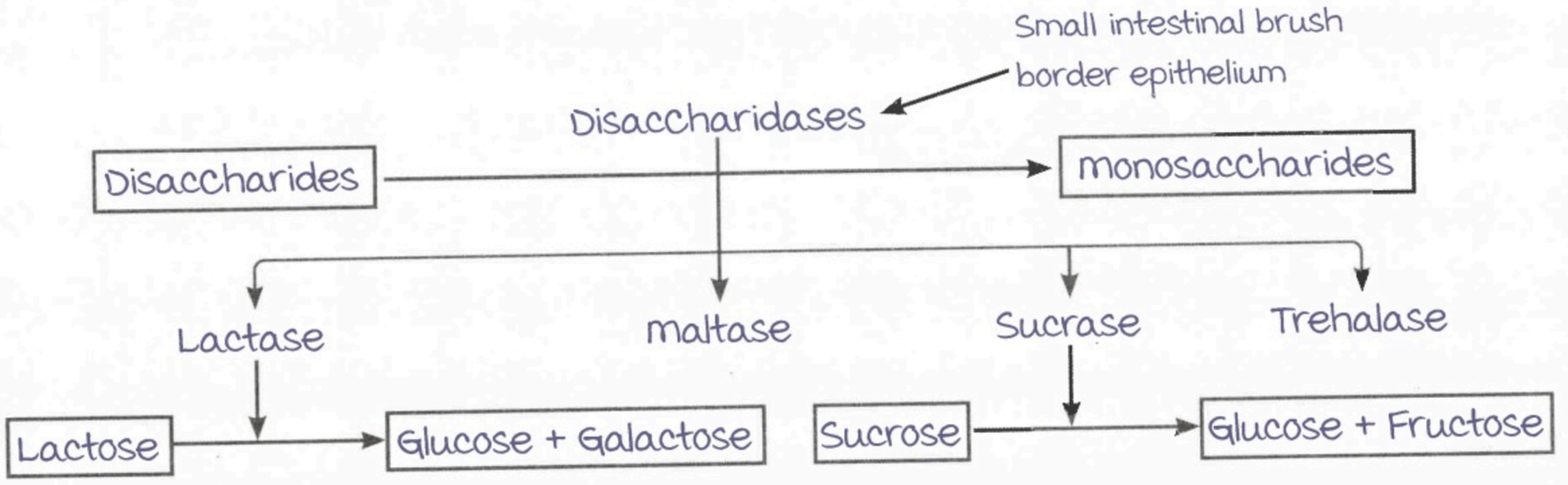
TYPES OF FATTY ACIDS

	Long chain	medium chain	Short chain
Carbon chain length	>12	8-12	<8
Origin	Large quantity in diet as triglycerides	Small quantity in diet (Coconut oil)	Bacterial degradation of unabsorbed carbohydrates in colon.
Primary absorption site	Small intestine	Small intestine	Colon
Require pancreatic lipolysis	Yes	No	No
Require micelle formation	Yes	No	No
Present in stool	Minimal	No	Substantial

Carbohydrate Absorption

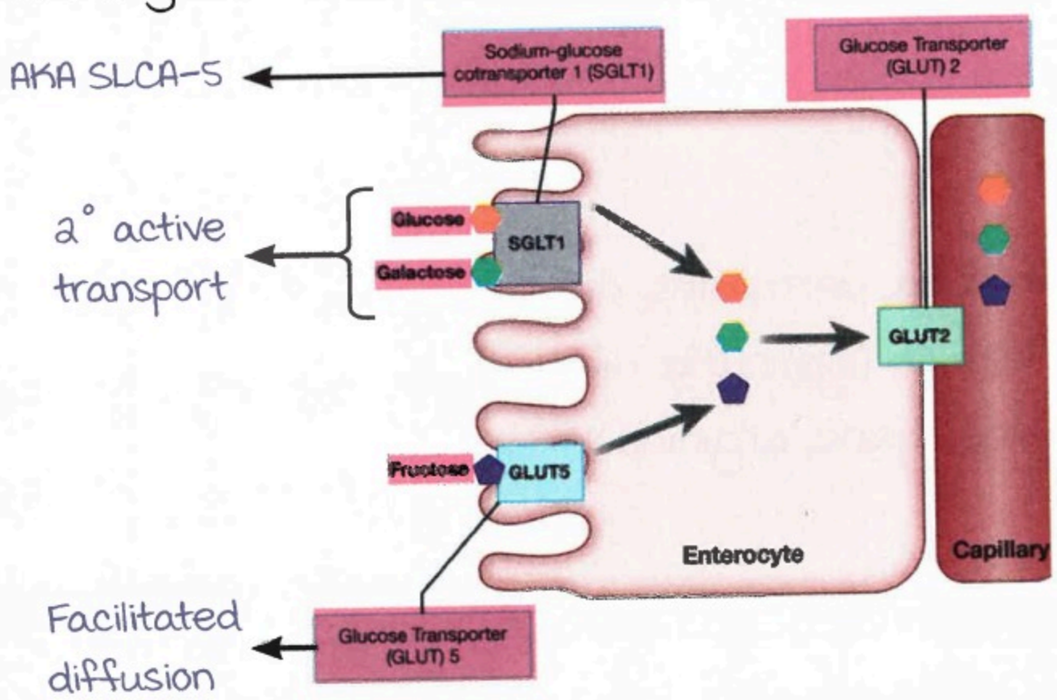
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MECHANISM OF CARBOHYDRATE ABSORPTION

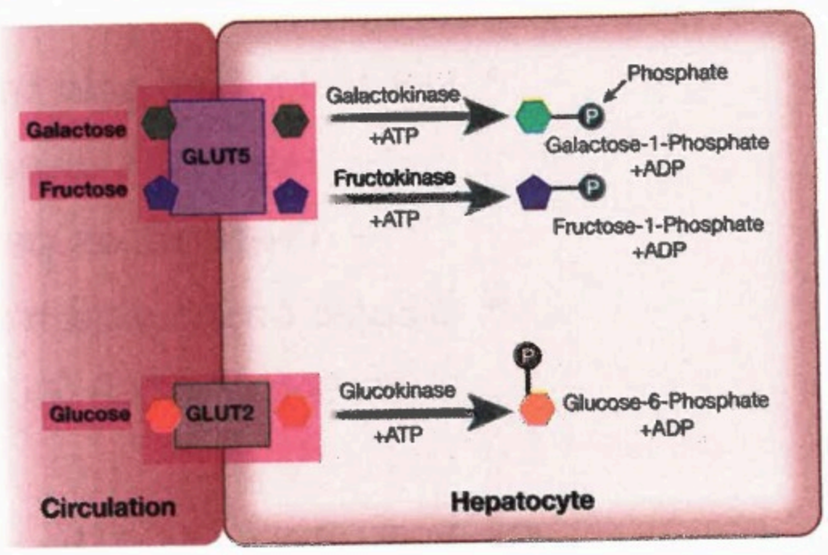


Receptors for absorption :

In the gut :



In the liver :



Note :

SGLT-1 defect → Familial glucose-galactose malabsorption syndrome.

DEFECTS IN CARBOHYDRATE ABSORPTION

Pathology	Etiology	malabsorption
Lactase deficiency	• Congenital : Autosomal recessive. • Acquired : – Primary. – Secondary.	Selective.
Small intestinal mucosal defect	• Celiac disease. • Whipple's disease. • Tropical sprue.	Global.

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Proteins

00:34:00

Mechanism of absorption :

Endopeptidase : Enterokinase (Produced by enterocyte brush border).

Trypsinogen → Trypsin

Trypsin → Activate chymotrypsin, elastase.

Exopeptidase : Carboxypeptidase and aminopeptidase (Activated by trypsin).

Transport in small intestinal mucosa : Sodium-amino acid co-transporters.

Defects in protein absorption :

Global malabsorption : Celiac disease, Whipple disease, Tropical sprue.

Selective malabsorption :

- Enterokinase deficiency.
- Neutral amino acid transport defect :
  - Hartnup's disease.
  - Presents as pellagra : Diarrhoea, dermatitis, dementia.
- Dibasic amino acid transport defect → Cystinuria → Excretion of COLA (Cystine, ornithine, lysine, arginine).

Carbohydrate Malabsorption

00:39:45

LACTOSE INTOLERANCE (LACTASE DEFICIENCY)

Physiology :

Lactose (Dietary : milk, cheese)

Lactase | Brush border

Glucose + Galactose

Etiology :

Congenital : Autosomal recessive (Absent enzyme from birth).

Accquired : ↓enzyme activity.

- Primary (m/c) : Genetic defect (allele persistence).
- Secondary :
  - Infections.
  - Irritable Bowel Syndrome (IBS).

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Pathogenesis :

Undigested carbohydrate

↓

↑ Unabsorbed carbohydrates in colon

↓ Fermented by colonic bacteria :
- Bacteroides fragilis
- Bifido bacteria
- Lactobacillus

↑ Acid and gases (H_2, methane, CO_2)

Clinical presentation :

- 90 min post prandial : Abdominal distension, bloating, cramps.
- Diaper dermatitis :
  - In children d/t acidic pH.
  - Severe watery diarrhea.
  - Failure to thrive.
  - Irritability.



Diaper dermatitis

Investigations :

1. Stool pH : Acidic.
2. Lactose hydrogen breath test :
   - Principle : Excess lactose in colon $\xrightarrow[\text{by bacteria}]{\text{Fermented}}$ methane + H_2 ⟶ Absorbed ⟶ Tested on breath.
   - Positive test : H_2 > 20 ppm.
3. Lactose-Glucose blood test : Obsolete.

management :

- Dietary management.
- Ca^{2+} supplementation.

Note :

Protein malabsorption :
- Hartnups disease : Similar to pellagra.
- Cystinuria : Excretion of COLA (Cystine, ornithine, lysine, arginine).

Fat Malabsorption

00:44:23

Post mucosal absorption disorders :

1. Abetalipoproteinemia.
2. Intestinal lymphangiectasia.

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ABETALIPOPROTIENEMIA

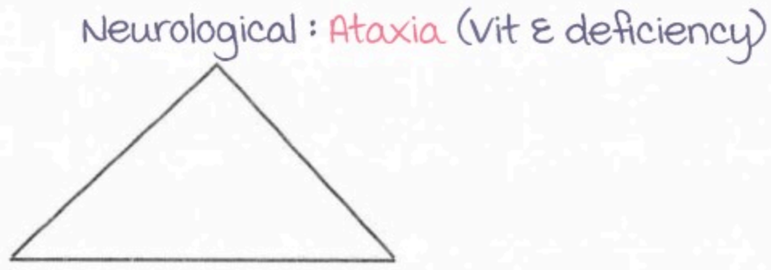
AKA Neuroacanthocytosis.

Etiopathogenesis :

- Autosomal recessive.
- Defect in MTTP (microsomal Triglyceride Transport Protein) → Defect in apoB48 → Defective esterification → No chylomicron formation.

Clinical presentation :

- Hypolipidemia.
- Triad of :



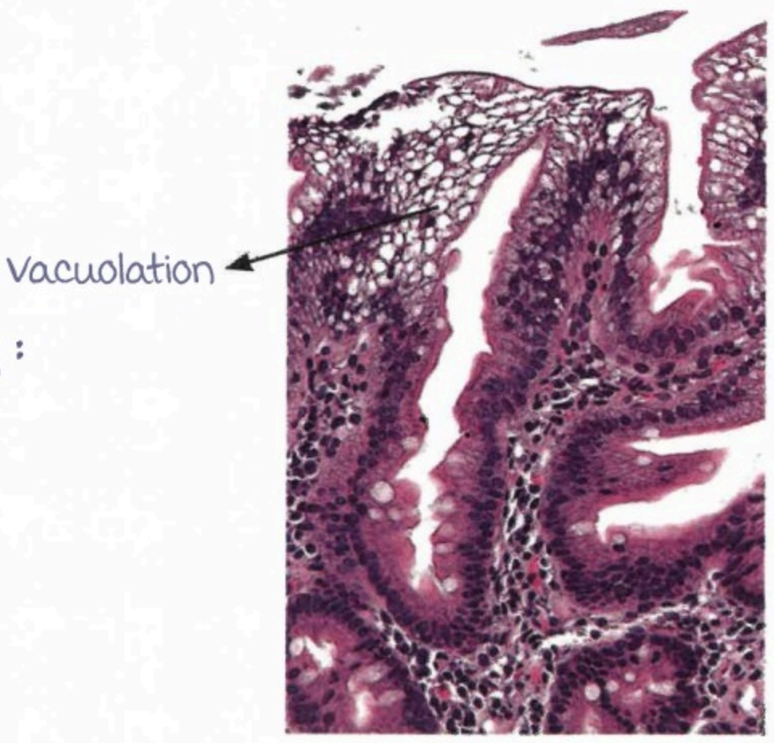
Investigations :

1. Peripheral Smear : Acanthocytes.
2. Post prandial endoscopic biopsy :
   - 100% diagnostic.
   - Vacuolation in enterocytes.

Note :

GI disorders which are 100% diagnostic on biopsy :

- Whipple's disease.
- Abetalipoproteinemia.
- Agammaglobulinemia.

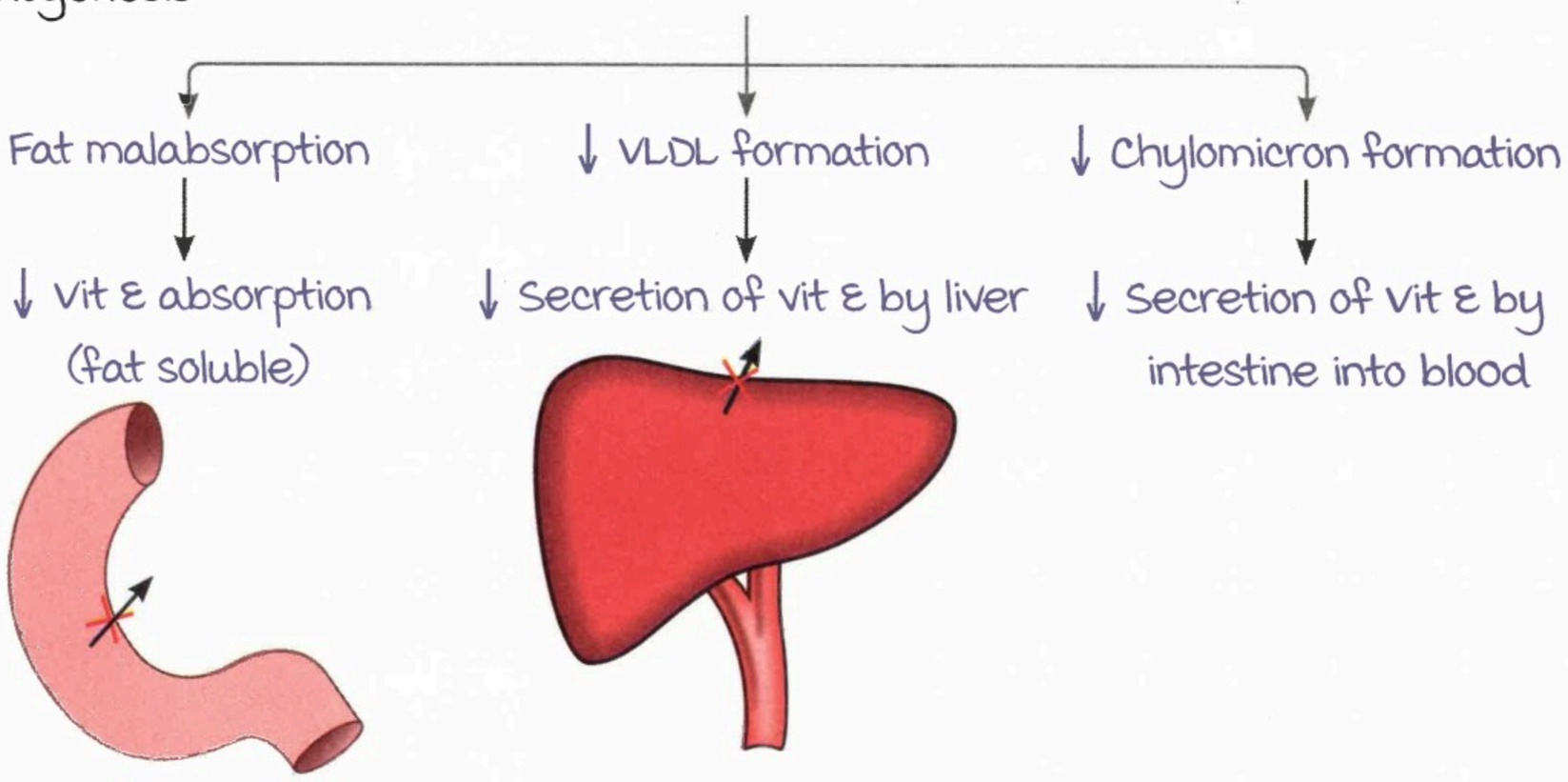


Duodenal biopsy of Abetalipoproteinemia

Vit E deficiency in Abetalipoproteinemia :

Clinical features : Ataxia (Pronounced).

Pathogenesis :



INTESTINAL LYMPHANGIECTASIA

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Physiology :

Chylomicrons —Across intestinal lymphatics→ Blood.

Etiopathogenesis :

Secondary : Lymph node obstruction → ↑ Lymphatic pressure → Exudation.

- Lymphoma.
- Crohn's disease.

Primary : Abnormally dilated lymphatic channel

↓ ↑pressure

Exudation into intestinal lumen

Clinical presentation :

1. Protein losing enteropathy :
   - most important : Primary intestinal lymphangiectasis.
   - Hypoalbuminemia with hypoglobulinemia.
   - Edema.
   - Ascites.

D/d :

- Nephrotic syndrome : Loss of albumin in urine.
- Liver disease : ↓albumin production.

2. Diarrhea :

Investigation :

- Fecal α-1 antitrypsin : Elevated (specific test).
- CT scan : Wall thickening of small intestine.
- Video capsule endoscopy.

Management :

Medium chain triglyceride diet.

